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Continuous Functions

1.1 Continuity of a Function

Definition 1.1.1

Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is said to be **continuous** if for each open subset V of Y , the set $f^{-1}(V)$ is an open subset of X .

Theorem 1.1.2

Let X and Y be topological spaces; let $f : X \rightarrow Y$. Then the following are equivalent:

1. f is continuous.
2. For every subset A of X , one has $f(\bar{A}) \subset \overline{f(A)}$.
3. For every closed set B of Y , the set $f^{-1}(B)$ is closed in X .
4. For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Remark.

If the condition in (4) holds for the point x of X , we say that f is **continuous at the point x** .

1.2 Homeomorphisms

Definition 1.2.1

Let X and Y be topological spaces; let $f : X \rightarrow Y$ be a bijection. If both the function f and the inverse function

$$f^{-1} : Y \rightarrow X$$

are continuous, then f is called a **homeomorphism**.